



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design) Course Project - An Intelligent Navigation UI

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An Intelligent Navigation UI

- Map For UI

- Download the map image file “MapForUI.jpg” that is 803x1404 pixels
- A pixel in the map image represents 1.7 meters in the real world
- Take the left-bottom corner as the origin of the real-world coordinates system
- Only roads in the map are navigable



- Intelligent Vehicles (IVs) For UI

- Each IV is represented by a rectangle of size 8 meters by 3 meters
- The visualization size of each IV is adjustable according to certain scaling factor

An Intelligent Navigation UI

- Pedagogical Awareness
 - The project is purely for pedagogical purpose rather than practical application
- Hard Requirements
 - Only in Matlab
 - Implementation in other languages is NOT accepted !
 - Main script that boots the UI is necessary
 - Do not need to do any ad hoc configuration
 - Can launch the UI simply by running the main script
 - Afterwards, all relevant operations can and should be done via the UI only
 - Minimum usage of built-in functions
 - Usage of basic mathematical built-in functions is allowed
 - Usage of basic image I/O (imread/imshow) and Matlab UI controls is allowed
 - Usage of others is NOT accepted !



It is not the result but the **observable process** that matters

An Intelligent Navigation UI

- Basic UI Requirements
 - Can launch the UI simply by running the main script, after which the Matlab command line interface should no longer be involved in following operations
 - Can load the map for visualization
 - Whenever clicking on a point in the map, display its real-world position (namely its coordinates)
 - Can load an IV (or multiple IVs) into the map
 - The UI should provide a way to initiate loading of a new IV
 - Load the IV by clicking on a point (namely the loaded point) in the map
 - The IV should be loaded onto roads which are the only navigable part in the map and hence there should be a mechanism to check whether the loaded point is valid or not
 - If the loaded point is invalid, an error message should be given
 - If the loaded point is valid, the map with the newly-added IV as well as already-existing IVs is visualized

An Intelligent Navigation UI

- Basic UI Requirements
 - IV loading with adjustable orientation
 - The UI should provide a way of adjusting the IV orientation besides above basic requirements
 - Can remove an IV from the map
 - The UI should provide a way to remove an unneeded IV while keeping other IVs not touched
 - Re-visualize the map after update of the IVs
 - Can report real-world positions of all loaded IVs
 - Can display the real-world distance of two clicked points in the map
 - Can display the real-world length of a trajectory characterized by a sequence of continuously clicked point
 - Can rotate the map
 - The UI should provide a way to specify a degree of rotation
 - The map rotated by the specified degree is visualized

An Intelligent Navigation UI

- Optional UI Requirements
 - Distributed Assignment
 - Each student takes one optional requirement and this part is scored independently
 - OR 1: Road skeleton extraction
 - The UI provide a way to manually extract the road skeleton (partially)
 - The UI can visualize the extracted road skeleton according to its real-world configuration
 - The UI can visualize only the (roughly) road part attached to the extracted road skeleton in the map
 - OR 2: Scale and local visualization
 - The UI can adjust the visualization size of the IV according to a scaling factor adjustable
 - The UI can visualize only the local map of the IV according to a range adjustable
 - The local map is a circle centered at the IV and its radius is the specified range
 - The visualized local map should conform to the specified range strictly

An Intelligent Navigation UI

- Optional UI Requirements

- OR 3: IV loading with orientation aligned automatically

- Besides previous basic requirements of IV loading, the IV orientation should be aligned automatically with the road
 - Can visualize the map together with the IV in certain rotated way as if the IV is always heading upward

- OR 4: Virtual street view generation

- Can generate the virtual street view associated with an arbitrary road point as if it was captured at that point

- OR 5: Path planning

- Given two arbitrary map points, can find their closest road points respectively
 - Can find the shortest road path between the two road points and visualize the shortest road path on the map

An Intelligent Navigation UI

- Submission Requirements
 - All Matlab scripts
 - Packaged into one zip file
 - Technical report
 - Describing your idea of UID
 - Describing how to use all the aspects of the UI
 - Describing how optional requirements are assigned to group members respectively
- Submission by email
 - Only one student of the group (say the group leader) sends an email to Prof. Li
 - The email should contain the packaged Matlab scripts as well as the technical report
 - The email should give a complete list of all group members
- Attention on Prof. Li's acknowledgement of reception

DISCUSSION

Q & A